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Title

BlockChain and D.L.T. Notes

Professor:
Massimo la Morgia

Student:
Carlo Da Roma

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1 Introduction

1.1 Definition and Fundamentals of Blockchain

A **blockchain** is a tamper-evident and tamper-resistant digital ledger implemented in a distributed architecture without requiring a central repository or a central governing authority (such as a central bank, corporation, or government).

At its core, a blockchain allows a network of untrusted participants to record transactions in a shared ledger. Under normal operation within a peer-to-peer network, once a transaction is verified and added to the blockchain, it becomes immutable and cannot be altered or deleted.

1.2 Evolution of Value Transfer Mechanisms

To understand why decentralized digital currencies are necessary, it is essential to trace how human trade systems evolved over time.

1.2.1 Barter Economy

Before money existed, trade relied on direct barter—exchanging goods directly for other goods.

- **Mechanism:** Alice trades medicine directly with Bob for a tool.
- **Limitation (Double Coincidence of Wants):** Trade can only occur if Alice wants what Bob has AND Bob wants what Alice has. If Alice needs a tool, Bob needs food, and Carol has food but needs medicine, direct trade fails unless all parties coordinate a multi-way swap simultaneously.

1.2.2 Credit-Based Systems

Credit systems introduced deferred settlement to bypass immediate coincidence of wants.

- **Mechanism:** Bob gives Alice a tool today, and Alice incurs a debt to provide goods to Bob at a future date.
- **Limitations:**
  - **Valuation Disagreements:** Assumes all goods or debts have agreed-upon equivalent value.
  - **Counterparty Risk:** High risk of default if one party refuses or is unable to settle the debt later.

1.2.3 Physical Cash Systems

Physical currency solves trade coordination and credit risk.

- **Advantages:** Trades can happen in any sequential order, goods can be assigned granular prices, transactions can be performed offline, and users maintain a high degree of privacy.
- **Disadvantages:** Systems require bootstrapping, physical transport of cash, and reliance on a central sovereign authority (e.g., government or central bank) to guarantee and back the token value.

1.2.4 Online Electronic Payments

Modern electronic payment infrastructure (e.g., credit cards, bank transfers, digital wallets) digitizes fiat currency.

- **Limitations:**
  - **Intermediary Reliance:** Requires trusted third parties (banks, payment processors) that collect fees.
  - **Lack of Anonymity/Privacy:** Identity and financial histories are fully accessible to intermediaries and merchants.
  - **Transaction Reversibility:** Transactions can be disputed or subjected to chargebacks, introducing settlement delay.

1.3 The Digital Cash Problem and Double-Spending

Creating a native digital cash system presents a fundamental computer science challenge: **the double-spending problem**.

Unlike physical banknotes, digital data can be duplicated flawlessly and infinitely at near-zero cost. If a virtual note is represented merely as a file, a user can copy-paste it and spend the exact same unit of currency multiple times.

Attempted solutions and their drawbacks:

1. **Digital Signatures by Issuers:** The issuer digitally signs virtual notes. While this verifies authenticity, it strips away anonymity and requires complete trust in the issuer.
2. **Unique Serial Numbers & Centralized Ledgers:** Each virtual note is assigned a unique identifier tracked in a centralized ledger. Every transaction must be validated against the central issuer to ensure the note hasn't been spent already.

1.3.1 Drawbacks of Centralized Authorities

Centralized payment models introduce critical vulnerabilities:

- **Transaction Censorship:** The central entity can unilaterally block or freeze accounts and transactions.
- **High Intermediary Fees:** Operational costs and profit margins of central entities increase transaction costs.
- **Single Point of Failure:** Central servers are vulnerable to outages, cyberattacks, and systemic corruption.
- **Reversibility:** Central entities can arbitrarily roll back or alter past balances.

1.4 Ledger Architectures and System Classifications

To evaluate ledger designs, it is necessary to distinguish between system topology and authority distribution.

Centralized: A single central node or authority processes, stores, and controls all data and state updates.

Decentralized: Decision-making and governance are distributed across multiple independent entities without a single controlling authority.

Distributed: Processing and data storage are shared across a network of interconnected physical computers.

Blockchains are structurally architected as distributed systems and politically structured as decentralized networks.

1.5 Theoretical Challenges in Distributed Systems

Building consensus across a distributed network is difficult due to fundamental limits summarized by Leslie Lamport: *"A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable."*

In a distributed environment, nodes cannot make any reliable assumptions regarding:

- **Network Timing:** Messages can be delayed, reordered, duplicated, or lost unpredictably.
- **Network Reliability:** Communication links may fail or experience intermittent partitions.
- **Node Availability:** Individual participant machines may crash, disconnect, or act maliciously at any time.

- **Global State:** No single participant ever possesses an instantaneous, perfectly synchronized view of the complete global network state.
- **Clock Synchronization:** Physical hardware clocks drift over time; true global time synchronization without trusted hardware is impossible.

1.6 Bitcoin: Origin and Key Milestones

Bitcoin introduced the first practical solution to the double-spending problem without requiring a trusted central authority, combining cryptographic hashing, public-key cryptography, peer-to-peer networks, and Proof-of-Work consensus.

- **August 18, 2008:** The whitepaper titled *"Bitcoin: A Peer-to-Peer Electronic Cash System"* was published under the pseudonym Satoshi Nakamoto.
- **January 3, 2009:** The Genesis Block (Block #0) was mined. Satoshi embedded a headline from *The Times* newspaper into the genesis block's coinbase parameter: *... "The Times 03/Jan/2009 Chancellor on brink of second bailout for banks"*. This served both as a timestamp and as a comment on central banking instability.
- **January 12, 2009:** The first Bitcoin transaction occurred between Satoshi Nakamoto and Hal Finney.
- **May 22, 2010 (Bitcoin Pizza Day):** Laszlo Hanyecz completed the first documented commercial purchase using Bitcoin, paying 10,000 BTC for two pizzas.

1.7 Cryptographic Assets: Coins, Altcoins, and Tokens

The broad cryptocurrency market consists of different types of digital assets:

- **Native Coins:** The native currency built directly into a underlying, standalone blockchain protocol (e.g., BTC on Bitcoin, ETH on Ethereum, SOL on Solana). Native coins pay for transaction execution (gas fees) and secure the consensus protocol.
- **Altcoins:** A general term for alternative cryptocurrencies created after Bitcoin.
- **Tokens:** Digital assets issued on top of existing smart contract-enabled blockchains (e.g., ERC-20 tokens on Ethereum). Tokens rely on the underlying blockchain's infrastructure and consensus mechanism for security.

1.7.1 Market Capitalization

In financial analysis, an asset's market size is evaluated using Market Capitalization:

$$\text{Market Capitalization} = \text{Current Unit Price} \times \text{Circulating Supply} \tag{1}$$

1.8 Non-Fungible Tokens (NFTs) and Decentralized Finance (DeFi)

1.8.1 Non-Fungible Tokens (NFTs)

While coins and standard tokens are *fungible* (interchangeable, where 1 BTC equals any other 1 BTC), Non-Fungible Tokens represent unique, non-interchangeable digital cryptographic identifiers on a blockchain that establish verifiable ownership of specific digital assets or rights.

1.8.2 Decentralized Finance (DeFi)

DeFi refers to an ecosystem of permissionless financial services (such as automated market makers, lending pools, decentralized exchanges, and yield protocols) deployed as smart contracts on distributed ledgers, removing traditional banking intermediaries.

1.8.3 Common Ecosystem Risks and Malicious Practices

- **Rug Pull:** Malicious developers attract capital into a token project or DeFi protocol, then abruptly withdraw all liquidity and abandon the project.
- **Wash Trading:** A form of market manipulation where an entity simultaneously buys and sells the same asset to fake trading volume and artificially inflate demand.
- **Pump and Dump:** Organized groups coordinate asset purchases to artificially drive up the price, then dump their holdings onto unsuspecting buyers.

1.9 Blockchain Decision Framework

A common misconception is that a blockchain is necessary for all multi-user digital applications. In reality, blockchains introduce high resource overhead and low throughput compared to traditional centralized databases.

To determine whether a application genuinely requires a blockchain solution, decision models (such as the Wüst & Gervais framework) evaluate system constraints sequentially:

1. **Do you need to store state?**
    - *No:* No database or blockchain is required.
    - *Yes:* Proceed to step 2.
2. **Are there multiple writers (entities modifying data)?**
    - *No:* Use a standard centralized database.
    - *Yes:* Proceed to step 3.

3. **Can you use an always-online Trusted Third Party (TTP)?**
    - *Yes:* Use a trusted centralized or replicated database managed by the TTP.
    - *No:* Proceed to step 4.
4. **Are all writers known?**
    - *No:* Use a **Permissionless Blockchain** (e.g., Bitcoin, Ethereum).
    - *Yes:* Proceed to step 5.
5. **Are all writers trusted?**
    - *Yes:* Use a shared database with access controls.
    - *No:* Proceed to step 6.
6. **Is public verifiability required?**
    - *Yes:* Use a **Public Permissioned Blockchain**.
    - *No:* Use a **Private Permissioned Blockchain**.

2 bitcoin

2.1 Introduction and Ledger Fundamentals

Bitcoin is a decentralized peer-to-peer (P2P) network designed to maintain a synchronized, distributed ledger across independent participants without relying on a central authority.

- **Ledger Properties:** The fundamental goal of the network is to ensure that all participating nodes maintain an identical copy of the state data (the ledger).
- **Immutability and Append-Only Model:** Any participant can submit new entries to be appended to the ledger, but no participant has the permission or ability to modify or delete existing historical data.

2.2 Transactions and Digital Signatures

A transaction is the fundamental record added to the ledger, representing a transfer of funds between account addresses.

- **Transaction Definition:** A transaction records the transfer of X coins from account A to account B.
- **Digital Signatures:** Every transaction must be cryptographically signed by the owner of the source funds (the sender). Digital signatures serve as the cryptographic analogue to physical handwritten signatures and satisfy two core security properties:

1. *Unforgeability:* Only the owner possessing the secret private key can produce a valid signature, yet anyone with access to the corresponding public key can easily verify its validity.
2. *Document Binding:* A digital signature is mathematically tied to a specific transaction payload. It cannot be detached and reused to endorse a different transaction.

- **Paper Check Metaphor:** A Bitcoin transaction functions similarly to a paper check. No monetary value changes hands until the transaction is submitted to the network, verified against available funds, and permanently settled onto the shared ledger.

2.3 Transaction Propagation and the Mempool

When a user initiates a transaction, it is propagated across the distributed network for validation and inclusion.

- **Peer-to-Peer Broadcasting:** A client forwards a new transaction to a node, which then broadcasts it to all adjacent nodes across the P2P network.
- **Mempool (Memory Pool):** Each node maintains its own independent mempool—a temporary holding area in memory for unconfirmed transactions that have passed basic network rules but have not yet been packed into a block.
- **Ordering Challenge:** Because multiple users submit transactions asynchronously across the globe, nodes must rely on consensus mechanisms to decide transaction ordering and block inclusion.

2.4 Distributed Systems Constraints

Operating a global consensus ledger over an untrusted network introduces inherent distributed systems challenges. As stated by Leslie Lamport, a distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable. Key constraints include:

- **Timing Delays:** Messages can experience variable delays, reordering, or loss across network routes.
- **Network Reliability:** Communication links can fail or recover unpredictably.
- **Node Availability:** Participating machines or processes may crash, disconnect, or restart at any time.
- **Absence of Global State:** No single node possesses a perfectly real-time, global view of the entire network state.

- **Clock Synchronization:** Physical hardware clocks drift, preventing reliance on global synchronized time.

2.5 Transaction Fees and Economics

Recording transactions on the blockchain incurs cost; senders must include a transaction fee to incentivize processing.

- **Fee Structure:** Fees are not calculated based on the monetary value (BTC amount) transferred, but rather on the physical data size of the transaction measured in bytes.
- **Fee Rate (sat/byte):** The fee rate is measured in Satoshis per byte (sat/B).
- **Competitive Fee Market:** Block space is strictly limited. During network congestion, users participate in an auction-style mechanism, offering higher fee rates to prioritize their transactions.
- **Mempool Prioritization:** Transactions offering higher sat/B are prioritized by miners for inclusion in the next block. Low-fee transactions risk remaining stuck in node mempools for extended periods.

2.6 Block Structure and Node Classification

2.6.1 Block Structure

Transactions are packed into fixed-size data blocks. The blockchain forms a cryptographic sequence of blocks sequentially linked to one another, where each block stores a reference hash (prev_hash) pointing to the immediately preceding block header.

2.6.2 Bitcoin Node Types

The network relies on two primary categories of full node architecture:

- **Client (Full) Nodes:**
  - Store a complete copy of the Bitcoin blockchain history.
  - Independently validate all transactions and blocks against protocol consensus rules.
  - Relay valid blocks and transactions to peers, supporting decentralization and fault tolerance.
- **Mining Nodes:**
  - Specialized full nodes that collect unconfirmed transactions and compete to solve Proof-of-Work cryptographic puzzles.

- Receive block rewards and collected transaction fees upon successful block discovery.
- Require specialized high-performance hardware (ASIC miners).

2.7 Distributed Consensus

In a network of n nodes where a subset may be faulty or malicious, a distributed consensus protocol must satisfy two core requirements:

1. **Agreement:** All honest nodes must agree on the exact same state value.
2. **Validity:** The value agreed upon must originate from an honest node.

2.7.1 Simplified Bitcoin Consensus Steps

1. New transactions are broadcast to all nodes in the network.
2. Each node collects unconfirmed transactions into a candidate block.
3. In each round, a random node (determined by Proof-of-Work) earns the right to broadcast its block.
4. Other nodes accept the block only if all contained transactions are valid (unspent outputs, valid digital signatures).
5. Nodes signal acceptance of the block by including its hash as the prev_hash reference in the next block they attempt to mine.

2.8 Cryptographic Hash Functions

A hash function is a deterministic mathematical function mapping binary strings of arbitrary length to fixed-length output strings known as hash values or digests.

- **Standard Algorithm:** Bitcoin uses the double SHA-256 cryptographic hash function.
- **Key Characteristics:** Computationally efficient to compute, collision-resistant, and sensitive to minute input changes (avalanche effect).

2.9 Proof-of-Work (PoW)

2.9.1 Conceptual Background

The concept originated as an anti-spam mechanism (e.g., Hashcash) designed to combat spam emails and Denial-of-Service attacks by requiring senders to expend verifiable CPU computational work.

2.9.2 Proof-of-Work in Bitcoin

To propose a new block, miners must solve a cryptographic puzzle requiring brute-force evaluation of a nonce parameter:

$$H(\text{nonce} \parallel \text{prev_hash} \parallel \text{tx}_1 \parallel \text{tx}_2 \parallel \cdots \parallel \text{tx}_n) < \text{Target} \tag{2}$$

The miner who succeeds in satisfying this condition earns the right to propose and broadcast the block to the network. Receiving nodes validate the proof and append the block to their local chain.

2.10 Rounds and Difficulty Adjustment

A block acts as the discrete state progress unit (or round) of the blockchain.

- **Target Block Time:** The network aims to maintain an average discovery time of 10 minutes per block.
- **Difficulty Recalibration:** To adapt to fluctuations in total network hashing power (miners joining or leaving), the target difficulty is automatically adjusted every 2016 blocks:

$$\frac{2016 \text{ blocks} \times 10 \text{ minutes}}{24 \text{ hours / day}} = 14 \text{ days (2 weeks)} \tag{3}$$

- **Genesis Block:** The initial block (height 0) from which the entire chain originates.

2.11 Incentives and Bitcoin Supply

Miners are economically motivated through block rewards and fees:

- **Coinbase Transaction:** The initial transaction created in each block by the miner to mint new Bitcoins and collect transaction fees.
- **Block Reward:** The subsidy of newly minted Bitcoins awarded per block (currently 3.125 BTC per block following periodic halving events).
- **Max Supply Limit:** The total maximum supply of Bitcoin is capped at $21,000,000$ BTC.

2.12 Forks and the Longest Chain Rule

2.12.1 Occurrence of Forks

If two or more miners solve the Proof-of-Work puzzle nearly simultaneously, multiple competing candidate blocks are broadcast. As a result, network nodes transiently adopt differing local versions of the blockchain, creating a fork.

2.12.2 Longest Chain (Cumulative Work) Rule

To maintain state convergence across the network:

- Nodes follow the canonical chain rule: selecting the valid chain possessing the greatest cumulative Proof-of-Work (commonly called the longest chain).
- Eventually, one fork branch grows faster than the other, leading all honest nodes to switch to the dominant canonical chain.
- Blocks on discarded branches become orphan/stale blocks, and any non-duplicate transactions in them return to the mempool for future processing.

References

[1] Leslie Lamport. *LaTeX: a Document Preparation System.* Addison Wesley, Massachusetts, 2 edition, 1994.